AELTS4017
LANGUAGE TRENDS THROUGH STATISTICS

Course Description

This course introduces students to the fundamentals of quantitative research in Second Language Acquisition (SLA) by integrating statistical methods with key topics in language learning. Students will explore research questions in SLA (e.g., the effects of instruction type, language anxiety, and age of acquisition on L2 learning) through hands-on data analysis using JASP. By the end of the course, students will be able to critically evaluate empirical studies, apply appropriate statistical methods (e.g., t-tests, ANOVA, correlation, regression, and chi-squared tests), and conduct their own quantitative analyses of SLA-related data.

** It is recommended that you take this course after 영어학개론 and/or 제 2 언어습득.

Note: As with any course, this syllabus provides a general description of the purpose, methods, content, and requirements of the course; specifics may be modified during the semester in response to class needs.

Course Objectives

By the end of the course, students will be able to:

- **Understand the role of quantitative research in SLA** and how statistical methods help answer key research questions.
- **Identify and measure linguistic variables** (e.g., age, motivation, input, bilingualism) and distinguish between different types of data.
- **Apply descriptive and inferential statistical techniques** (e.g., correlations, t-tests, ANOVA, regression) to analyze language learning data.
- **Use JASP software** to conduct statistical analyses and interpret results, and G*Power to conduct power analyses.
- **Evaluate empirical studies in SLA**, recognizing potential biases, research design limitations, and the validity of conclusions.
- **Design and conduct their own small-scale quantitative study** related to SLA and present findings in a group presentation and a group video.

Course Textbook

- Loerts, H., Lowie, W., & Seton, B. (2020). *Essential Statistics for Applied Linguistics: Using R or JASP* (second edition). Bloomsbury Publishing.
- Other required readings, videos, datasets, and handouts can be found on the E-campus website (주차학습).

Course Expectations

- **Late Work Policy:** Late work will only be accepted without penalty in cases of documented personal emergencies (e.g., illness, family emergencies) or officially excused situations (e.g., university-approved events). Students must communicate with the professor as soon as possible regarding any extenuating circumstances. The acceptance of late work and any accommodations will be at the professor's discretion, based on the nature of the situation and supporting documentation.

- **Communication through E-Campus:** All important announcements get sent out via E-campus, and it is your responsibility to check E-campus and your email account regularly (at least once a day). Failure to check E-campus regularly will not be considered a valid excuse for failing to complete an assignment or for failing to do so according to specified directions. All course materials get uploaded to E-campus, so this is where you want to check if you miss a session and need to catch up on class work.

- **Collaboration with Peers:** This course relies heavily on collaboration with peers. As you engage with statistical analyses and SLA research, you will work in designated groups to explore data, discuss findings, and support each other in the learning process.

    - Group Seating & Collaboration – You will sit in groups and regularly collaborate on data analysis exercises and discussions.
    - Peer Support for Technical Issues – If you encounter technical difficulties (e.g., using JASP, interpreting statistical results), seek help from your group members before reaching out to the instructor.

- **Personal laptop required:** Students are required to bring a personal laptop to each class session, with each group having at least 2–3 laptops available for collaborative data analysis. This course involves hands-on work using JASP and other statistical tools, so laptops should be fully charged and capable of running JASP smoothly.

- **Academic Integrity:** Academic integrity is essential in this course. All work must be your own original creation, tailored to course requirements. While AI tools (e.g., ChatGPT) may assist with brainstorming, writing, or data analysis, students

must critically engage with and refine AI-generated content. Directly submitting AI-generated work without proper attribution or modification is considered plagiarism. Violations may result in a zero on the assignment.

Course Requirements

The general requirements for this course are that students complete all assigned readings and assignments, learn, and contribute to the learning of others. Students should actively engage with and respond to all methods of instruction, including in-class discussions. Specific course requirements/assignments are below, detailed guidelines will be uploaded to E-campus.

- **Attendance and Participation (10%):** You are expected to come to all class sessions on time, having read all assigned readings for the day and prepared to take part in various activities/discussions we will do in class. If you absolutely cannot avoid being absent, you are required to inform the professor ahead of time. You are responsible for class materials missed due to any absence or tardiness. The professor may, but is not obliged to, permit a student to make up for missed hours/work. Six or more absences will result in automatic failure of the course. Three tardies (arriving 5 minutes after class begins) equal one absence. For each session, you can earn 1 point for attendance and 1 point for participation.

- **Weekly Assignments (25%):** Weekly quizzes and exercises are a **core component of this course**, designed to reinforce both **SLA concepts and statistical methods**. These assignments will include:
    - **Reading Quizzes (RQs)** – Short multiple-choice quizzes on the assigned readings, given at the beginning of class (RQ #1–#9). Students who arrive late miss the quiz; an RQ missed due to an excused absence must be made up before the next session.
    - **JASP Assignments** – Individual JASP homework (e.g., JASP Practicals 4A, 4B/4C, and 5B, submitted with HW templates) and in-class JASP workshops applying statistical methods to SLA datasets.
    - **Practice Sets and Group Work** – Stats Practice sets, handouts, and group assignments that reinforce key statistical concepts.

- **Group Presentations (15%):** There will be two group projects: (1) Group Presentation 1 (Week 7), an in-class presentation of a small-scale study based on data collected from at least 15 participants, and (2) the Final Group Video Project (due in Week 15). Detailed guidelines will be uploaded to E-campus.

- **Midterm (25%):** The midterm will assess students' understanding of research methods and statistical analyses covered in the course (Weeks 1-7). The exam consists of 30 items (multiple-choice and short-response questions) and will include:

  - **Conceptual Questions** – Testing knowledge of research design and statistical reasoning.
  - **Data Interpretation** – Analyzing datasets and interpreting statistical results.
  - **Application-Based Problems** – Selecting and justifying appropriate statistical methods for SLA studies.

- **Final Exam (25%):** The final exam will cover materials from Weeks 9–14 and will follow the same format as the midterm.

- **Extra Credit:** There will be several opportunities for you to earn extra credit during the semester.

Grading

Semester grades in this course are **criterion-referenced** and **individual-referenced**. This means that your grade is based on the extent to which you meet the course learning objectives and demonstrate individual progress, rather than being curved against other students' performance. Our course follows the University's official grading system:

Grade	Percentage range	Grade	Percentage range
A+	97-100	C0	74-76
A0	94-96	C–	70-73
A–	90-93	D+	67-69
B+	87-89	D0	64-66
B0	84-86	D–	60-63
B–	80-83	F	56-59
C+	77-79		

The course grade is based on the following weightings:

Requirements	%
Attendance and Participation	10%
Weekly Assignments	25%
Group Presentations	15%
Midterm	25%
Final Exam	25%

Course Schedule (Tentative)

Week	Date	Topics, Readings & Assignments
Unit 1: Foundations of Measurement and Descriptive Statistics		
1	3/1 (Mon)	*No class (National holiday: Independence Movement Day)*
	3/3 (Wed)	**Topic: Course Introduction** - Why statistics in SLA research? Statistical literacy and quantitative research - Key purposes of quantitative research: data reduction, inference, discovery of relationships - Four steps of research; turning abstract constructs into measurable data - Course overview, expectations, and group assignment **Recommended Readings** - Loerts et al. (2020) Chapter 1 **Assignments** - Complete the pre-course survey before the first class - Download and install JASP on your laptop
2	3/8 (Mon)	**Topic: Constructs and Variables** - Constructs and operationalization - Types of variables (categorical vs. continuous) - Validity of measurement **Required Readings** - Loerts et al. (2020) Chapter 2 (pp. 19-26): Complete all activities in the chapter - Loerts et al. (2020) JASP Practical 1 (pp. 197-202): Follow all steps in JASP (Data-Practical1.csv) **Assignments** - Reading Quiz (RQ) #1 - Handout 01: Some Preliminaries of Statistics
	3/10 (Wed)	**Topic: Descriptive Statistics (1)** - Central tendency (mean, median, mode) - Dispersion (range, IQR, standard deviation) and z-scores **Required Readings** - Loerts et al. (2020) Chapter 3 (pp. 27-42; Section 3.4 may be skipped): Complete all activities - Loerts et al. (2020) JASP Practical 2 (pp. 203-207) **Assignments** - RQ #2 - Handout 01 (p. 2): Bring the completed page to class - JASP Practice with Quiz 1 Dataset (individual; due Sunday, 11:59 PM)
3	3/15 (Mon)	**Topic: Descriptive Statistics (2)** - Frequency distributions (ungrouped vs. grouped) and their visualization (histogram, bar graph, line graph) - Shapes of distributions: normal, skewed, bimodal; kurtosis - Relationships between two continuous variables: scatterplots **Required Readings**

Week	Date	Topics, Readings & Assignments
		- No new reading: Review Loerts et al. (2020) Chapter 3 carefully
	3/17 (Wed)	**Topic: More Preliminaries to Statistics** - Sample size and frequency distributions - Sample vs. population; statistic vs. parameter; estimation - Research procedure: RQ, operationalization, scales, functions of variables (IV vs. DV) **Assignments** - In-class JASP Workshop: Descriptive Statistics (group; due Friday)
Unit 2: Statistical Logic: Getting Ready for Inferential Statistics		
4	3/22 (Mon)	**Topic: Statistical Logic (1)** - Functions of variables: independent, dependent, and control variables - Sampling and representativeness; random sampling vs. biased sampling - Sampling error **Required Readings** - Loerts et al. (2020) Chapter 4 (pp. 43-50, until 4.2.1 "Hypotheses"): Read at least twice - Loerts et al. (2020) JASP Practical 3 Part A (#1-4, pp. 208-209) **Assignments** - RQ #3 - Group homework (handout p. 1)
	3/24 (Wed)	**Topic: Statistical Logic (2)** - Standard error (SE) and the sampling distribution - Null vs. alternative hypotheses - Steps in hypothesis testing; significance level (α) **Required Readings** - Loerts et al. (2020) Chapter 4 (pp. 51-55, until 4.5 "Degrees of freedom")
5	3/29 (Mon)	**Topic: Statistical Logic (3)** - The meaning of the p-value - Type I (α) vs. Type II (β) errors - One-tailed vs. two-tailed testing **Required Readings** - Review Loerts et al. (2020) Chapter 4 through p. 55 (Section 4.4) - Required videos on E-campus: standard error; hypothesis testing **Assignments** - Stats Practice-1 (due Tuesday, 11:59 PM)

Week	Date	Topics, Readings & Assignments
	3/31 (Wed)	**Topic: Statistical Logic (4)** - Test statistic vs. p-value - Assumptions (normality, homogeneity of variance); parametric vs. non-parametric tests **Required Readings** - Finish Loerts et al. (2020) Chapter 4 (Section 4.8 may be skipped) - Required video on E-campus: parametric vs. non-parametric tests **Assignments** - RQ #4 - Group Assignment: Statistical Logic: Power & Effect Size
6	4/5 (Mon)	**Topic: Statistical Logic (5)** - Statistical power and Type II error - Effect size - Power analysis (G*Power) - Statistical significance vs. practical significance **Assignments** - Group handout discussion (group speakers share responses) - Review the Q&A slides on hypothesis testing (included on the midterm)
	4/7 (Wed)	**Topic: Statistical Logic (6)** - Effect size benchmarks (Cohen, 1988; Plonsky & Oswald, 2014) - Margin of error and 95% confidence intervals - Criticisms of null hypothesis significance testing (p-values) **Required Readings** - Required videos on E-campus (confidence intervals): Watch BEFORE reading Section 4.8 - Loerts et al. (2020) Section 4.8 **Assignments** - Stats Practice-2 - In-class JASP Workshop: Independent Samples T-Test (group; due Sunday, 11:59 PM)
7	4/12 (Mon)	**Topic: Statistical Logic (7) and Reporting Practices** - Interpreting confidence intervals (precision and direction) - Degrees of freedom - Choosing a statistical test; assumption checks (Shapiro-Wilk, Levene's test) - Reporting t-test results (independent samples t-test, Welch's t-test, Mann-Whitney U test)
	4/14 (Wed)	**Group Presentation 1** - Groups present a small-scale study based on data collected from at least 15 participants **Assignments** - Midterm Review Questions (posted on E-campus)
Midterm		
	4/19 (Mon)	**Midterm Exam (Weeks 1-7)**

Week	Date	Topics, Readings & Assignments
8	4/21 (Wed)	**Topic: Midterm Results Analyzed in JASP** - Descriptive and inferential analyses of the (anonymized) class midterm dataset
Unit 3: Comparing Groups and Exploring Relationships		
9	4/26 (Mon)	**Topic: Comparing Two Groups** - Independent samples t-test; Welch's t-test or Mann-Whitney U test - Paired samples t-test; Wilcoxon signed-rank test - One-sample t-test **Required Readings** - Loerts et al. (2020) Chapter 5: Sections 5.1 and 5.3 (skip 5.2, p. 79, and discussions of R) - Larson-Hall (2015) Chapter 5: Section 5.6 - Loerts et al. (2020) JASP Practical 3 (Data-Practical3a, 3b) **Assignments** - RQ #5 - Handout 07: Comparing Two Groups
	4/28 (Wed)	**Topic: Exploring Relationships (1)** - Pearson correlation: direction and strength - The correlation coefficient as an effect size - Correlation vs. causation **Required Readings** - Loerts et al. (2020) Chapter 5, pp. 63-68 - Larson-Hall (2015) Section 5.2 **Assignments** - RQ #6 - Handout 07 (Section II): Bring the completed handout to class - JASP Practical 4A (individual; HW template; due at the start of the next class)
10	5/3 (Mon)	**Topic: Exploring Relationships (2)** - Correlation practice in JASP (Flege et al., 1999 data) **Assignments** - JASP Practical 4A due
	5/5 (Wed)	*No class (National holiday: Children's Day)*
11	5/10 (Mon)	**Topic: Chi-Square Tests** - Chi-square tests for categorical data - JASP workshop (Labov sample data) **Required Readings** - Loerts et al. (2020) Section 5.2.3 (calculation parts may be skipped) - Larson-Hall (2015) Section 5.5 - Required video on E-campus: chi-square test **Assignments** - Handout 09: Chi-square tests - JASP Practical 4B and 4C (individual; HW template; submit as PDF; due Wednesday)
	5/12 (Wed)	**In-Class Activity: GrammarBot** - Group data analysis and interpretation using the GrammarBot

Week	Date	Topics, Readings & Assignments
		and AELTS language preference datasets **Assignments** - Post group interpretations to the GrammarBot Activity board (due the following day)
Unit 4: Regression		
12	5/17 (Mon)	**Topic: Regression Analysis (1)** - Simple linear regression: slope, intercept, residuals - The null hypothesis in regression **Required Readings** - Loerts et al. (2020) Sections 6.1 and 6.2 (skip discussions of R) - Required video on E-campus **Assignments** - RQ #7
	5/19 (Wed)	**Topic: Regression Analysis (2)** - JASP workshop: simple linear regression **Required Readings** - No new reading: Review Sections 6.1-6.2 **Assignments** - JASP Practical 5B (individual; HW template; due the following Monday) - Final Group Video Project guidelines released: Decide on your group's presentation format
13	5/24 (Mon)	**Topic: Regression Analysis (3)** - Multiple regression: R^2 vs. adjusted R^2 - Unstandardized b vs. standardized β - Assumptions and multicollinearity **Assignments** - RQ #8 - JASP Practical 5B due - In-class group work on the final video project
	5/26 (Wed)	**Topic: Regression Analysis (4)** - Multiple regression activity (handout) **Assignments** - In-class group work on the final video project
Unit 5: Comparing More than Two Groups		
14	5/31 (Mon)	**Topic: One-Way ANOVA** - The F-ratio: between-group vs. within-group variance - Why not multiple t-tests? (Type I error inflation) - Post hoc tests (e.g., Tukey's HSD) **Required Readings** - Loerts et al. (2020) Sections 7.1-7.2 **Assignments** - RQ #9 (the last RQ)
	6/2 (Wed)	**Topic: Factorial ANOVA** - Main effects and interaction effects **Required Readings**

Week	Date	Topics, Readings & Assignments
		- Loerts et al. (2020) Sections 7.3-7.5 **Recommended Readings** - Loerts et al. (2020) Chapter 8 **Assignments** - Handout 14: Factorial ANOVA
Wrap-Up		
15	6/7 (Mon)	**Final Review and Video Viewing** - Viewing of final group videos - Group-generated final review questions; Final Activity **Assignments** - Final Group Video due before class (10:00 AM) - Handout 14: Final Review
	6/9 (Wed)	**Optional Zoom Q&A Session**
16	6/14 (Mon)	**Final Exam (Weeks 9-14)**